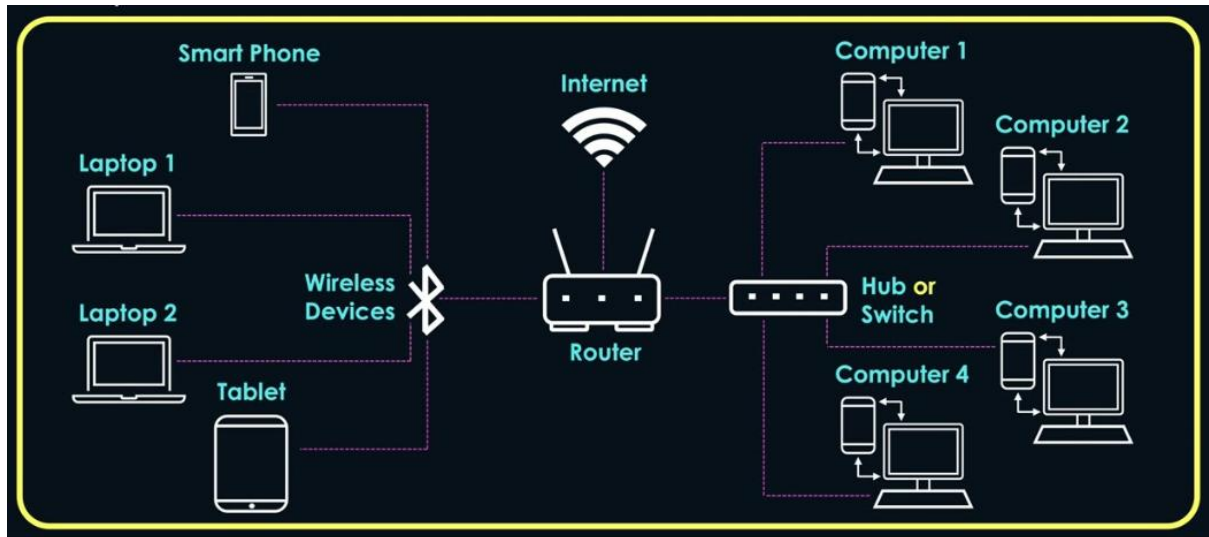


DAY-1

Tasks

1.

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2.

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Connects your home network to the Internet Service Provider (ISP) and converts the signal used by the ISP into a form your network can use.	Used at home to receive an Internet connection from an ISP.
Router	Connects different networks and directs data packets to the correct destination using IP addresses.	Used at home to share one Internet connection with phones, laptops and TVs.
Switch	Connects multiple wired devices on the same local network and forwards data to the appropriate device.	Used in an office to connect computers, printers and servers.
Access Point	Provides wireless/Wi-Fi access to devices and connects them to the wired network.	Used in a house, college or office to provide Wi-Fi coverage.

Device	Main Function	Real-Life Example
Network Cable	Provides a physical connection for transmitting network data between devices.	Ethernet cable used to connect a computer to a router or switch.

3.

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

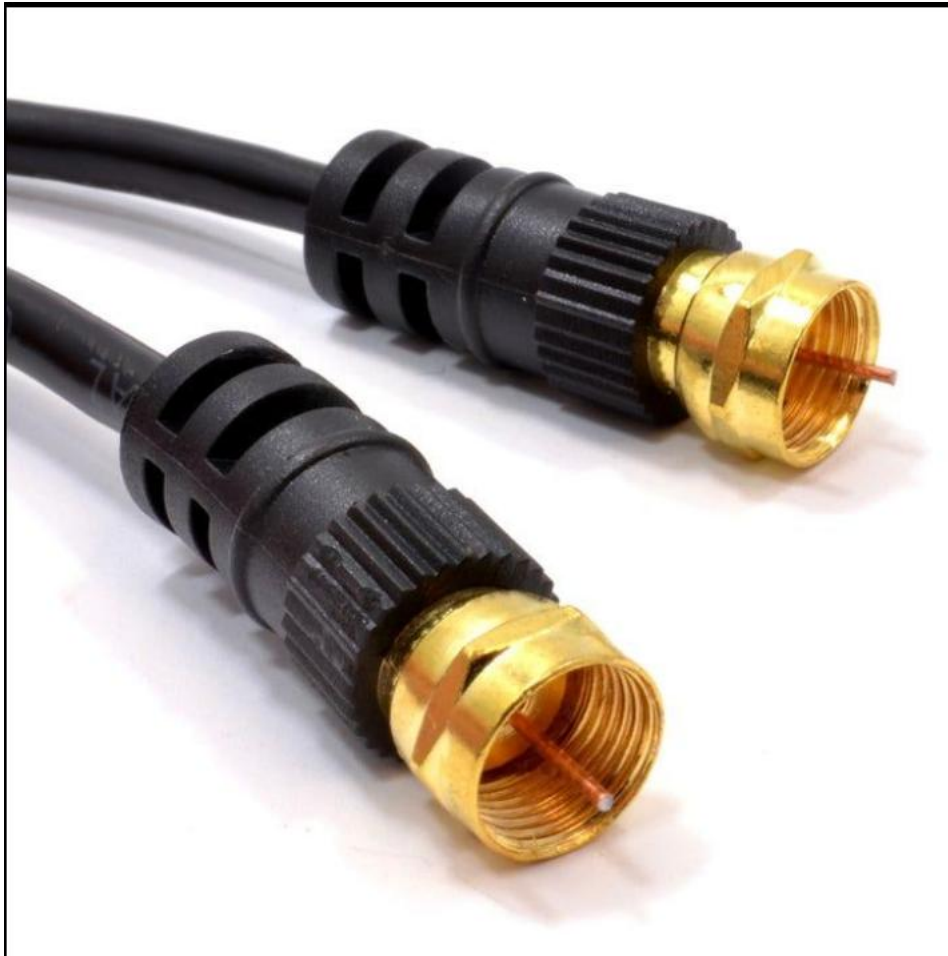
ETHERNET CABLE

Ethernet cables, commonly twisted-pair copper cables, are widely used to connect computers, routers, switches and other network devices in local networks.



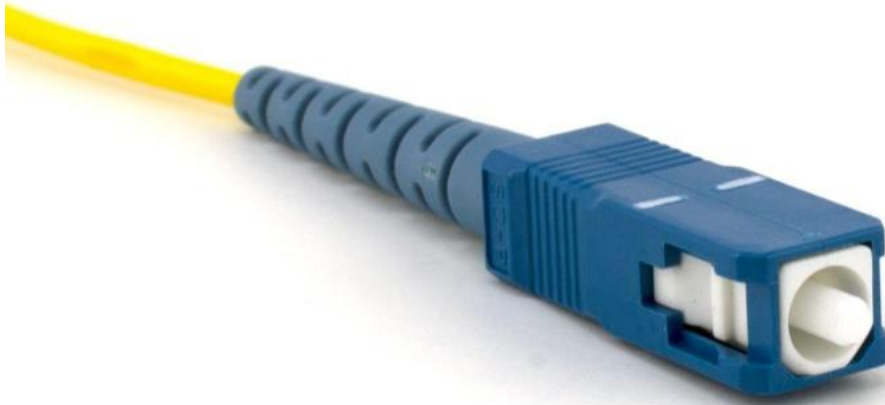
COAXIAL CABLE

Coaxial cables are commonly used for cable Internet connections, cable television and some older Ethernet networking systems.



FIBRE-OPTIC CABLE

Fiber-optic cables transmit data using light and are commonly used for high-speed Internet, telecommunications and long-distance network connections.



4.

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Difference	Router	Switch
Main use	Connects different networks and directs traffic between them.	Connects multiple devices within the same local network.
Typical location	Usually placed between the modem/ISP connection and the home or office network.	Usually placed inside a LAN to connect wired devices.
Data handling	Uses IP addresses to determine where packets should go.	Uses MAC addresses to forward frames to the correct device.
Network level	Primarily works at Layer 3 (Network Layer).	Primarily works at Layer 2 (Data Link Layer).
Internet connection	Can connect a local network to the Internet.	Normally does not directly connect separate networks to the Internet.

Difference	Router	Switch
Example	Home Wi-Fi router connecting your home network to the ISP.	Office switch connecting computers, printers and servers.
Wireless capability	Many home routers include Wi-Fi.	A basic switch normally provides wired connections only.
Traffic direction	Routes packets between different networks.	Forwards frames between devices on the same LAN.